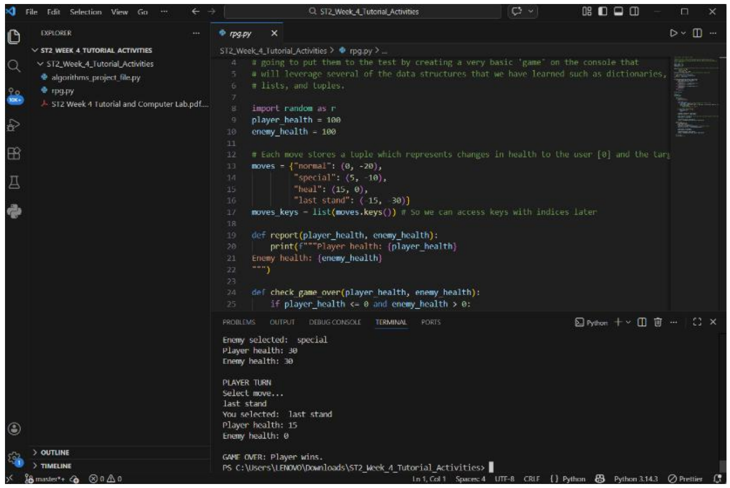
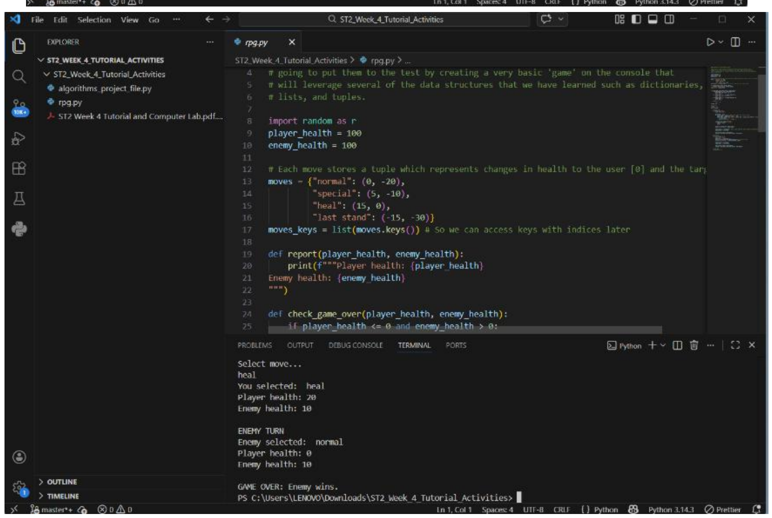
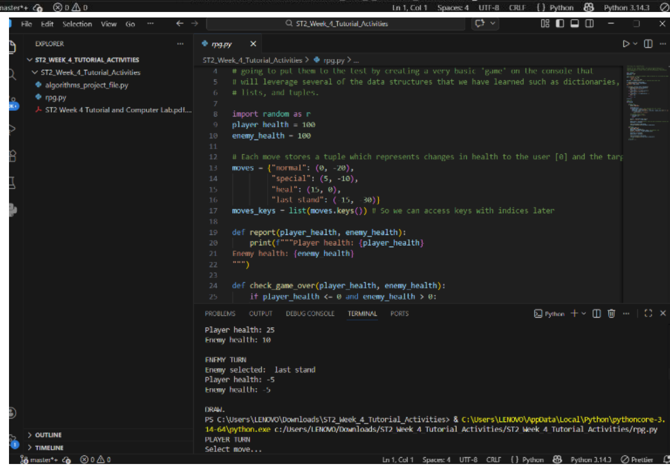
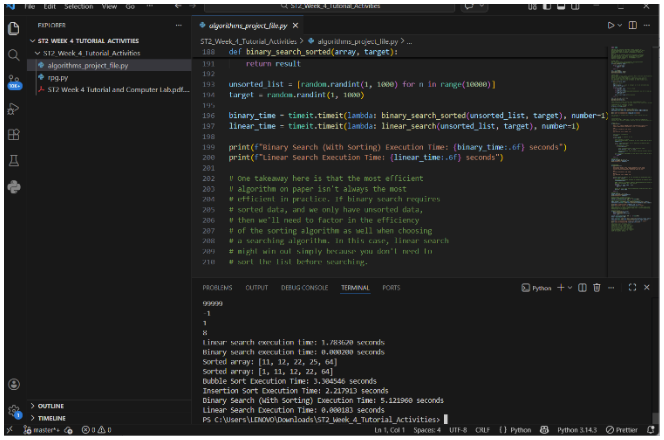
Task 1:Understanding Simple Data Structures in Python

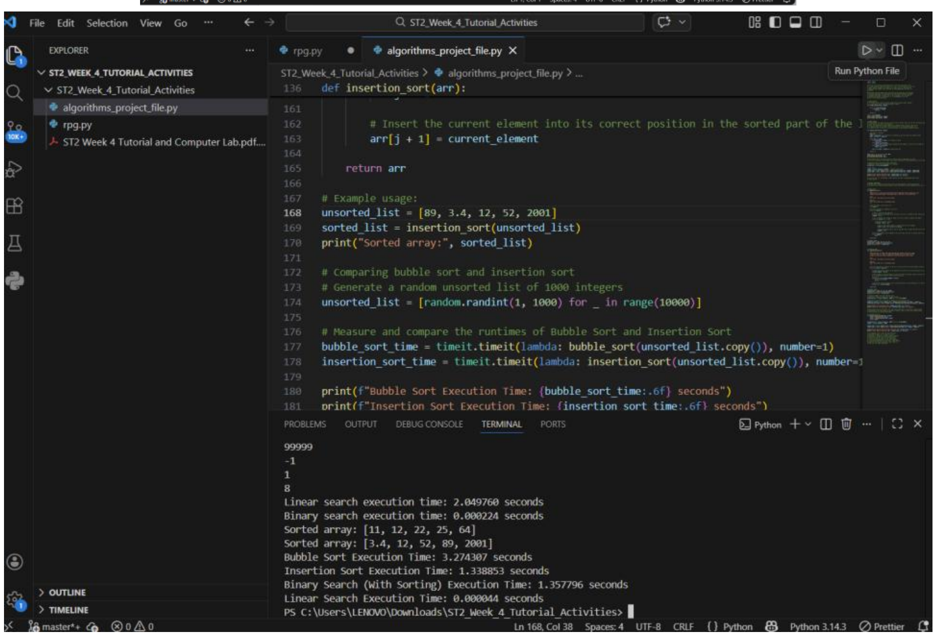




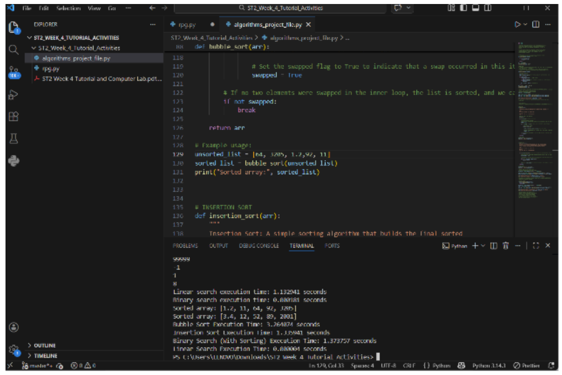


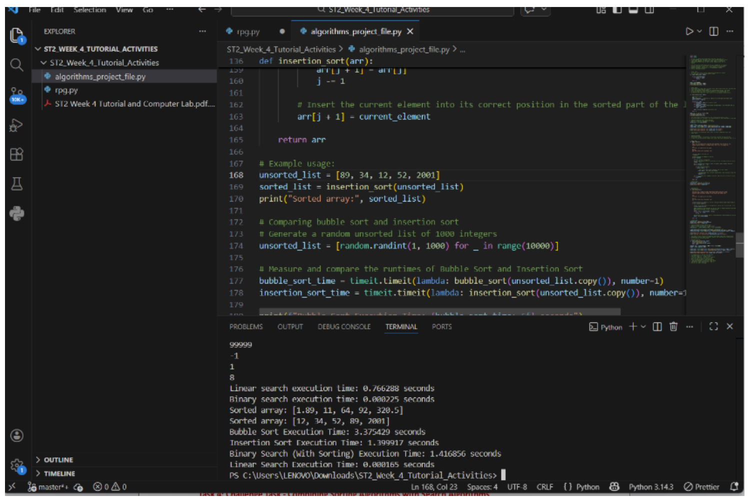
Task 2: Understanding Simple Search Algorithms and Measuring their Computational Efficiency in Python





Task 3: Understanding Simple Sorting Algorithms and Measuring their Computational Efficiency in Python





Task 4: Challenge Task - Combining Sorting Algorithms with Search Algorithms

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\LENOVO\AppData\Local\Programs\Microsoft VS Code> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/LENOVO/Desktop/ST2_Week_4_Tutorial_Activities (1)/ST2_Week_4_Tutorial_Activities/algorithms_project_file.py"

==============================
      DATA SIZE = 1000
==============================
Insertion Sort + Linear Search: 0.019800 sec
Bubble Sort + Linear Search:    0.057268 sec
Insertion Sort + Binary Search: 0.019488 sec
Bubble Sort + Binary Search:    0.044860 sec

==============================
      DATA SIZE = 3000
==============================
Insertion Sort + Linear Search: 0.195801 sec
Bubble Sort + Linear Search:    0.454742 sec
Insertion Sort + Binary Search: 0.199373 sec
Bubble Sort + Binary Search:    0.808435 sec

==============================
      DATA SIZE = 5000
==============================
Insertion Sort + Linear Search: 0.675477 sec
Bubble Sort + Linear Search:    1.474553 sec
Insertion Sort + Binary Search: 0.629988 sec
Bubble Sort + Binary Search:    1.285437 sec
PS C:\Users\LENOVO\AppData\Local\Programs\Microsoft VS Code>
```